

Exp No.11

## 1.Single Level Directory

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main()
```

$$\{$$

```
char file[20][20], fname[20];
```

```
int i, n = 0, choice, found;
```

```
while(1)
```

$$\{$$

```
printf("\n1. Create File");
```

```
printf("\n2. Delete File");
```

```
printf("\n3. Search File");
```

```
printf("\n4. Display Files");
```

```
printf("\n5. Exit");
```

```
printf("\nEnter your choice: ");
```

```
scanf("%d", &choice);
```

```
switch(choice)
```

$$\{$$

case 1:

```
printf("Enter file name: ");
```

```
scanf("%s", file[n]);
```

```
n++;
```

```
printf("File created successfully\n");
```

```
break;
```

case 2:

```
printf("Enter file name to delete: ");
```

```
scanf("%s", fname);
```

```
found = 0;
```

```
for(i = 0; i < n; i++)
```

$$\{$$

```
if(strcmp(fname, file[i]) == 0)
```

$$\{$$

```
for(int j = i; j < n-1; j++)
```

```
strcpy(file[j], file[j+1]);
```

```
        n--;  
        found = 1;  
        printf("File deleted\n");  
        break;  
    }  
}
```

```
if(found == 0)  
    printf("File not found\n");  
break;
```

case 3:

```
printf("Enter file name to search: ");  
scanf("%s", fname);  
found = 0;
```

```
for(i = 0; i < n; i++)  
{  
    if(strcmp(fname, file[i]) == 0)  
    {  
        printf("File found\n");  
        found = 1;  
        break;  
    }  
}
```

```
if(found == 0)  
    printf("File not found\n");  
break;
```

case 4:

```
printf("Files in directory:\n");  
for(i = 0; i < n; i++)  
    printf("%s\n", file[i]);  
break;
```

case 5:

```
return 0;
```

default:

```
printf("Invalid choice\n");
```

```
}
```

```
}
```

```
}
```

## 2.Compile and Run

```
gcc OS_exp11_1.c -o OS_exp11_1  
./OS_exp11_1
```

## 3.Expected output

```
1. Create File  
2. Delete File  
3. Search File  
4. Display Files  
5. Exit  
Enter your choice: 1  
Enter file name: file1  
File created successfully  
  
1. Create File  
2. Delete File  
3. Search File  
4. Display Files  
5. Exit  
Enter your choice: 1  
Enter file name: file2  
File created successfully
```

## 2.Mutli-Level Directory

### 1.Write the c program OS\_exp11\_2.c

```
#include <stdio.h>
#include <string.h>

struct directory
{
    char dname[20];
    char fname[20][20];
    int fcount;
};

int main()
{
    struct directory dir[10];
    int dcount = 0;
    int choice, i, j;
    char dirname[20], filename[20];

    while(1)
    {
        printf("\n1.Create Directory");
        printf("\n2.Create File");
        printf("\n3.Display");
        printf("\n4.Exit");
        printf("\nEnter choice: ");
        scanf("%d",&choice);

        switch(choice)
        {
            case 1:
                printf("Enter directory name: ");
                scanf("%s",dir[dcount].dname);
                dir[dcount].fcount = 0;
                dcount++;
                printf("Directory created\n");
                break;

            case 2:
                printf("Enter directory name: ");
                scanf("%s",dirname);

                for(i=0;i<dcount;i++)
                {
                    if(strcmp(dirname,dir[i].dname)==0)
                    {
                        printf("Enter file name: ");
                        scanf("%s",filename);

                        strcpy(dir[i].fname[dir[i].fcount],filename);
                        dir[i].fcount++;
                    }
                }
            }
        }
    }
```

```

        printf("File created\n");
        break;
    }
}
break;

case 3:
    printf("\nDirectory Structure:\n");

    for(i=0;i<dcount;i++)
    {
        printf("\n%s:\n",dir[i].dname);
        for(j=0;j<dir[i].fcount;j++)
        {
            printf("  %s\n",dir[i].fname[j]);
        }
    }
    break;

case 4:
    return 0;

default:
    printf("Invalid choice\n");
}
}
}

```

## 2.Compile and Run

```
gcc OS_exp11_2.c -o OS_exp11_2
./ OS_exp10_2
```

## 3.Expected output

```

1.Create Directory
2.Create File
3.Display
4.Exit
Enter choice: 1
Enter directory name: rahna
Directory created

1.Create Directory
2.Create File
3.Display
4.Exit
Enter choice: 1
Enter directory name: os
Directory created

1.Create Directory
2.Create File
3.Display
4.Exit

```

### 3.Sequential Allocation

#### 1. Write the c program OS\_exp11\_3.c

```
#include <stdio.h>

int main()
{
    int n, i, start, length;

    printf("Enter number of files: ");
    scanf("%d",&n);

    for(i=0;i<n;i++)
    {
        printf("\nEnter starting block and length of file %d: ", i+1);
        scanf("%d %d",&start,&length);

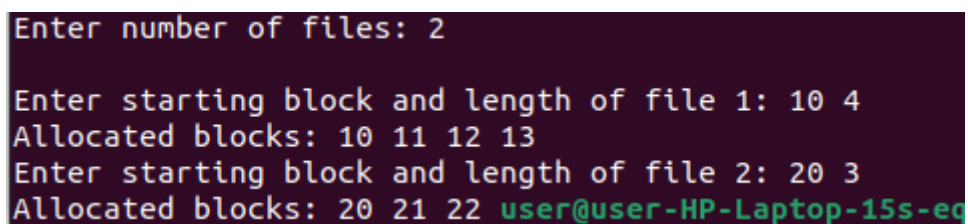
        printf("Allocated blocks: ");
        for(int j=start;j<start+length;j++)
        {
            printf("%d ",j);
        }
    }

    return 0;
}
```

#### 2.Compile and Run

```
gcc OS_exp11_3.c -o OS_exp11_3
./OS_exp11_3
```

#### 3.Expected output



```
Enter number of files: 2

Enter starting block and length of file 1: 10 4
Allocated blocks: 10 11 12 13
Enter starting block and length of file 2: 20 3
Allocated blocks: 20 21 22 user@user-HP-Laptop-15s-eq
```

#### 4.Linked Allocation

##### 1.Write the c program OS\_exp11\_4.c

```
#include <stdio.h>

int main()
{
    int f[50], i, j, n, start, len, k;

    for(i = 0; i < 50; i++)
        f[i] = 0;

    printf("Enter number of files: ");
    scanf("%d", &n);

    for(i = 0; i < n; i++)
    {
        printf("\nEnter starting block: ");
        scanf("%d", &start);

        if(f[start] == 0)
        {
            printf("Enter number of blocks needed: ");
            scanf("%d", &len);

            printf("Enter the blocks: ");

            for(j = 0; j < len; j++)
            {
                scanf("%d", &k);

                if(f[k] == 0)
                {
                    f[k] = 1;
                    printf("%d -> ", k);
                }
                else
                {
                    printf("Block already allocated\n");
                    break;
                }
            }

            f[start] = 1;
            printf("NULL\n");
        }
        else
        {
            printf("Starting block already allocated\n");
        }
    }
}
```

```
    return 0;  
}
```

## 2.Compile and Run

```
gcc OS_exp11_4.c -o OS_exp11_4  
./ OS_exp11_4
```

## 3.Expected output

```
Enter number of files: 1  
  
Enter starting block: 5  
Enter number of blocks needed: 4  
Enter the blocks: 7 10 12 18  
7 -> 10 -> 12 -> 18 -> NULL
```

## 5.Indexed Allocation

### 1.Write the c program OS\_exp11\_5.c

```
#include <stdio.h>
int main()
{
    int f[50], i, j, index, n, k;

    for(i = 0; i < 50; i++)
        f[i] = 0;

    printf("Enter index block: ");
    scanf("%d", &index);

    if(f[index] == 0)
    {
        printf("Enter number of blocks needed: ");
        scanf("%d", &n);
    }
    else
    {
        printf("Index block already allocated\n");
        return 0;
    }

    printf("Enter the blocks: ");

    for(i = 0; i < n; i++)
    {
        scanf("%d", &k);

        if(f[k] == 0)
        {
            f[k] = 1;
        }
        else
        {
            printf("Block already allocated\n");
            return 0;
        }
    }

    f[index] = 1;

    printf("File allocated successfully\n");

    printf("Index Block %d contains: ", index);

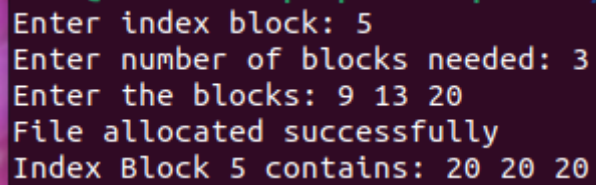
    for(i = 0; i < n; i++)
        printf("%d ", k);

    return 0;}
```

## 2.Compile and Run

```
gcc OS_exp11_5.c -o OS_exp11_5  
./OS_exp11_5
```

## 3.Expected output

A terminal window with a dark purple background and light green text. The text shows the program's execution flow: prompting for an index block, the number of blocks, the block values, a success message, and the final state of the index block.

```
Enter index block: 5  
Enter number of blocks needed: 3  
Enter the blocks: 9 13 20  
File allocated successfully  
Index Block 5 contains: 20 20 20
```

## Result

**File Organization Techniques and Allocation Strategies were implemented successfully**